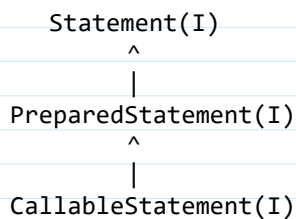


Statement vs PreparedStatement vs CallableStatement

09 August 2026 11:40

1. We can use normal Statement to execute multiple queries:
 - i. `st.executeQuery(query1);`
 - ii. `st.executeQuery(query2);`
 - iii. `st.executeQuery(query3);`i.e., if we want to work with multiple queries then we should go for Statement object.
2. If we want to with only one query, but should be executed multiple times then we should go for PreparedStatement.
3. If we want to work with stored procedure and functions then we should go for CallableStatement



`Statement st = con.createStatement();`
↑ this statement can execute any number of queries or multiple queries.

`PreparedStatement pst = con.prepareStatement(query);`
`pst.executeUpdate();`
⋮
`pst.executeUpdate();`
Advantage: the query will be compiled only once. So, performance will improve.

`CallableStatement st = con.prepareCall("{ call getEmpInfo(??) }");`
`st.execute();`



Need of Batch Update

- o Statement Batch updates.
- o PreparedStatement Batch updates.
- o CallableStatement Batch Updates.
- o Batch Updates and Transactions.

Batch Update

A batch update is a batch of updates grouped together and sent to the database in one "batch" rather than sending the updates one by one. So there is less network traffic involved in sending one batch of updates. (only one round trip) And the database might

be able to execute some of the updates in parallel. The speed up compared to the executing the updates one by one. You can batch SQL insert, update and deletes. It does not make sense to batch select statements.

There are three ways to execute batch updates:

1. Using a Statement
2. Using a PreparedStatement
3. Using a CallableStatement

Statement Batch updates

You can use a statement object to execute batch updates. You do so using the `addBatch()` and `executeBatch()` methods. Here is an example:

```
Statement st = con.createStatement();
st.addBatch("UPDATE employee set ename = 'Vijay' where emp_id = 'E020' ");
st.addBatch("UPDATE employee set ename = 'Ravi' where emp_id = 'E030' ");

int[] recordsAffected = st.executeBatch(); //
//Submits a batch of commands to the database for execution and if all commands execute
//successfully, returns an array of update counts.
```

From <<https://docs.oracle.com/javase/8/docs/api/java/sql/Statement.html>>

```
st.close();
```

PreparedStatement Batch Updates:

You can also use a PreparedStatement object to execute batch updates. The prepared statement enables you to reuse the same SQL statement and just insert a new parameters into it for each update to execute. For example:

```
String sql = "update employee set ename=?, basic_salary = ? where emp_id = ?";
```

```
PreparedStatement pst = con.prepareStatement(sql);
pst.setString(1, "Ram");
pst.setFloat(2, 2000.0);
pst.addBatch();
```

```
pst.setString(1 "Bharat");
pst.setFloat(2, 3000.0);
pst.addBatch();
```

```
int[] affectedRecords = pst.executeBatch();
```

```
pst.close();
```

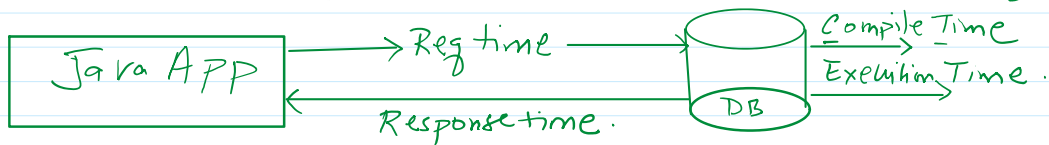
3. CallableStatement Batch Updates: It works similar to PreparedStatement Batch Updates

It is important to keep in mind that each update added to. A statement or prepared statement is executed separately by the database. That means if some of them may succeed before one of them fails, all the statements that have succeeded are now committed to the database, but the rest of the updates may not be. This can result in an inconsistent data in the database. To avoid this, you can execute the batch update inside a transaction. When executed inside a transaction, you can make sure that either all updates are executed or none. Any successful updates can be rolled back in case of update fails.

Need of batch updates

1. Statement:- When we submit multiple SQL queries to the database

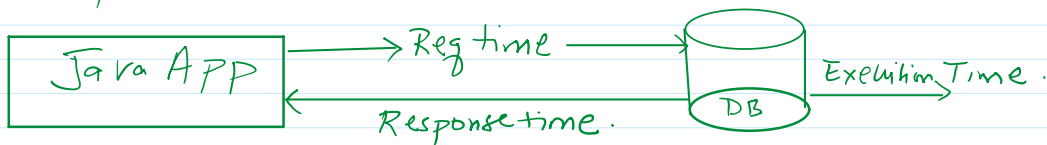
one by one then lot of time wasted in request and response.



$$\begin{aligned}\text{Time/query} &= \text{Reg. time} + \text{C.T.} + \text{E.T.} + \text{Resp.T.} \\ &= 1\text{ms} + 1\text{ms} + 1\text{ms} + 1\text{ms} = 4\text{ms}.\end{aligned}$$

$$\text{Time/1000 query} = (4 \times 1000) = 4000\text{ ms}$$

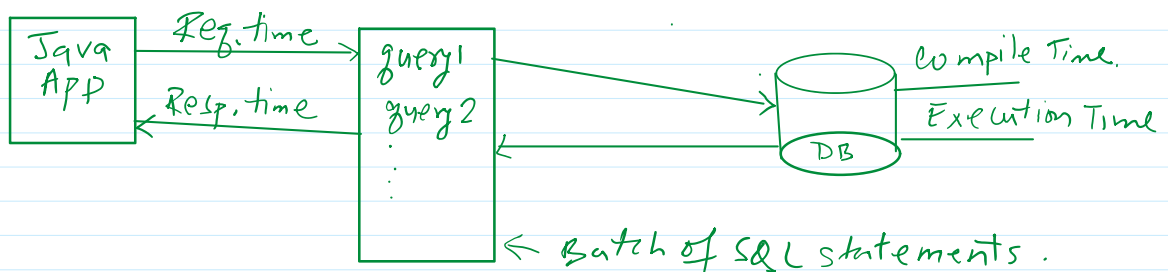
2. Prepared Statement:



$$\begin{aligned}\text{Time/query} &= \text{Reg. time} + \text{E.T.} + \text{Resp.T.} \\ &= 1\text{ms} + 1\text{ms} + 1\text{ms} = 3\text{ms}.\end{aligned}$$

$$\text{Time/1000 query} = (3 \times 1000) = 3000\text{ ms}$$

To overcome these problems, we should go for batch update. We can group all related SQL queries in a single batch and we can send that batch at a time to database.



$$\begin{aligned}\text{Time per 1000 queries} &= \text{R.T} + 1000 \times \text{C.T.} + 1000 \times \text{ET} + \text{Resp.T.} \\ &= 1 + 1000 \times 1\text{ms} + 1000 \times 1 + 1\text{ms} = 2002\text{ ms}.\end{aligned}$$

Hence the main advantage of batch updates are:

1. We can reduce network traffic.
2. We can improve performance.

We can implement batch updates by using the following

We can implement batch updates by using the following two methods:

1. `public void addBatch (String sqlQuery)`
2. `int[] executeBatch()` ← to execute batch of sql queries.